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Palaeomagnetism of Permian red beds in the contact aureole of the Tertiary Adamello intrusion (northern Italy)

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Permian clastic red beds of the Verrucano Lombardo formation have been contact metamorphosed by Eocene–Oligocene calc-alkaline intrusions of the Adamello massif (Southern Alps). The magnetization of the unmetamorphosed or at most anchimetamorphosed sediments outside the contact zone is controlled by pigmentary and detrital haematite. The pigment haematite is converted to biotite in the contact zone, where magnetite and pyrrhotite are also generated in the zone of highest-grade metamorphism. The detrital haematite (martite and titanohaematite), however, survives even in close proximity to the intrusion, where temperatures around 600 °C have been reached.

An apparent paradox is found: the structure of natural remanent magnetization (NRM) becomes less complex with increasing chemical changes owing to contact metamorphism. Except for viscous components, the NRM of the unmetamorphosed red beds consists of two magnetizations: an early acquired Permian component and a later overprint of Tertiary age. Owing to the natural chemical demagnetization, however, which removed the pigmentary haematite, a simple one-component NRM is often found in the metasediments of the contact zone. This magnetization is either of Permian origin, residing in the original detrital haematite, or of Tertiary age. The latter palaeomagnetic component is carried mainly by secondary haematite formed during contact metamorphism. The Permian magnetization has partly survived even in the highest metamorphic (andalusite) zone.

1. Geological introduction

The calc-alkaline rocks of the Adamello massif constitute the largest Alpine intrusion (550 km^2) which originates from the Tertiary orogenic cycle. Recent geochronological results yield an Eocene–Oligocene age for the different parts of the intrusion (Del Moro et al., 1985). Major tectonic faults form the boundary of the intrusion: the Tonale line in the north and the Giudicarie line in the east (Fig. 1). Mainly towards the south and west, the area is covered by red sandstones of Middle–Upper Permian (possibly Lower Triassic) age, known as the 'Verrucano Lombardo' formation (Assereto and Casati, 1966).

These predominantly terrestrial sediments, which overlie discordantly the metamorphic crystalline basement, can be subdivided into three units. Lower Permian conglomerates at the base of the formation are followed by silts, clays and volcanics of the Collio formation, on top of which the Verrucano Lombardo Sandstone has been

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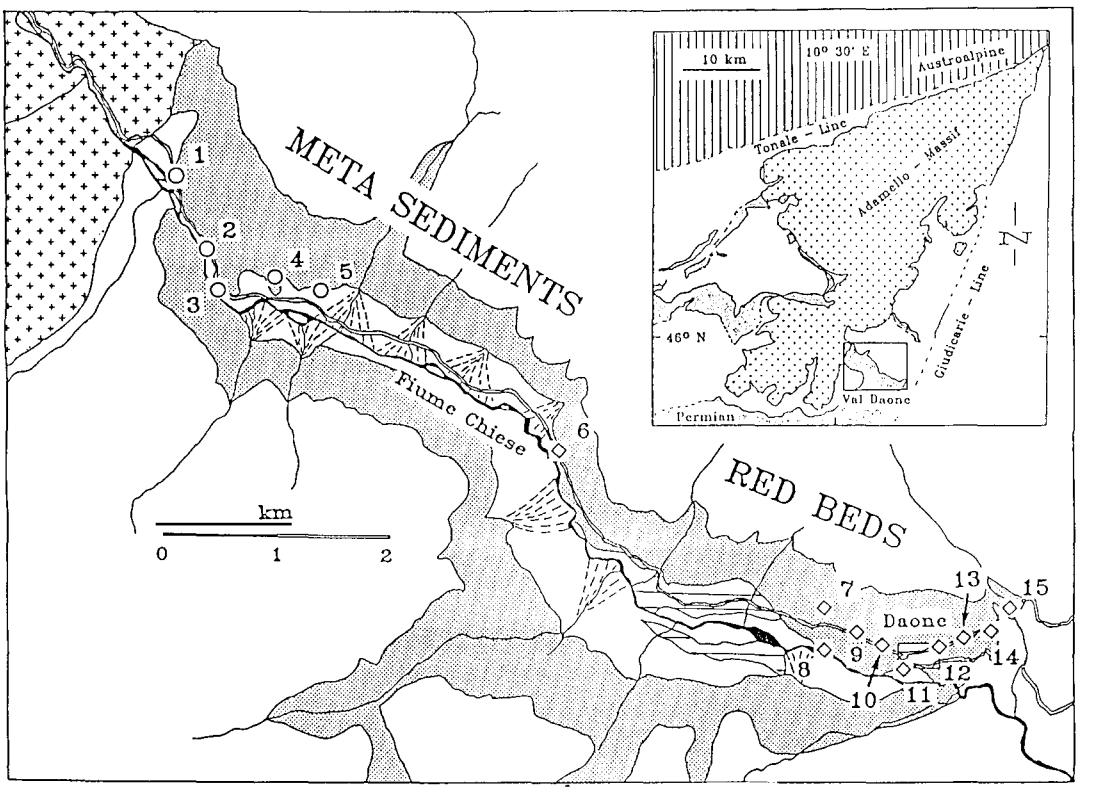


Fig. 1. Geological sketch map of the Adamello Massif. Crosses: Tertiary intrusives; stippled shading: Permian sediments; white: crystalline basement. Sites 6–15 in unmetamorphosed Permian red beds, sites 1–5 in metasandstones.

deposited (Cassinis and Peyronel-Pagliani, 1976). The unfolded and flat-lying red sediments reach a thickness of several hundred metres in the Val Daone (Fig. 1). They have been described as arkoses, lithic arkoses and lithic arenites (Riklin, 1983). Rock fragments, quartz (up to 80% by volume), plagioclase, K-feldspar and micas are embedded in the clay-sized matrix of these red and immature sandstones (Cassinis, 1968). The matrix consists mainly of clay (illite) minerals (55–85%), finely dispersed authigenic haematite (5–25%) and quartz (10–20%) (Riklin, 1983). The intensity of the red coloration directly depends on the volume proportion of the matrix haematite component: the siltstones have a strong red to blood-red coloration, whereas the coarse, conglomeratic layers show a reddish to yellow tint only. The visible Fe-oxide minerals have been identified in polished sections as (titano)haematite with exsolution lamellae of ilmenite or as haematite in the form of martite.

Structure and mineral paragenesis of the Permian red beds change completely in the closer proximity of the Adamello pluton owing to the increasing influence of contact metamorphism. Approaching the intrusion contact in the Val Daone (Fig. 1), the sandstones start to be metamorphosed at distances < 6 km from the intrusion contact. The red coloration darkens at first to deep red and red–violet. At distances < 2 km from the contact the sandstones become paler and end up as brownish grey quartzites. The most highly metamorphic sandstones (at the contact to the intrusion) are typical cordierite-spotted, white to grey hornfelses.

According to Riklin (1983) the matrix starts to recrystallize as a result of the contact metamorphism, and new micas are formed. The Fe needed is gained from the finely dispersed haematite of the rock matrix. Isochemical conditions control the metamorphic process. Reactions such as the

following are responsible for the mica production

ilmenite/titaniferous haematite + phengite

$$+ \text{K-feldspar} + H_2O \rightarrow \text{biotite} + \text{rutile} + \text{quartz}$$

Crossing the contact aureole the biotite metamorphic zone merges into the cordierite zone, which is finally superseded by the andalusite zone. The sedimentary texture can still be recognized. There is some evidence for sillimanite in the highest metamorphic zone (Riklin, 1983). Mineralogical temperature estimates (from the Si^{4+} content in muscovite) yield baking temperatures of $\sim 600\,^{\circ}C$ directly at the contact with the intrusion. The highest-grade metasandstones generally contain magnetite which apparently formed by a reduction process (Riklin, 1983). On the other hand, Fe^{3+} and Ti^{4+} have formed a new generation of (titano)haematite and rutile which has crystallized as small-sized grains in the matrix.

Since the contact metamorphism has affected the magnetic minerals of the red beds, their magnetic properties must also have been influenced. This raises questions about the palaeomagnetic significance of the rocks: to what extent has the original Permian magnetization of the sandstones been reset by the Tertiary metamorphic event? Second, how is the discoloration related to changes in the rock magnetic properties? Finally, can the unblocking characteristics of the natural remanent magnetization (NRM) be used to unravel the temperature history in the neighbourhood of the intrusion?

In order to investigate the palaeomagnetic and rock magnetic properties of the red beds, a palaeomagnetic profile was sampled along the Val Daone (Fig. 1). The Val Daone crosses radially the contact aureole in the southeast of the Adamello pluton. About 300 samples were drilled at 15 outcrops along a 10 km long profile. The location of the drilling sites was predetermined mainly by the partly inaccessible, glacially eroded valley topography. For this reason it was not possible to place the sites at regular intervals and to take samples from the same stratigraphic layers or even correlate the sites stratigraphically.

2. Rock magnetic properties

Coercivity spectra of isothermal remanent magnetization (IRM) and its thermal unblocking characteristics (Lowrie and Heller, 1982) as well as the temperature behaviour of strong field magnetization (M_s(T)) were studied in order to identify the ferromagnetic phases of the sandstones.

The rock magnetic properties of the red anchimetamorphosed to unmetamorphosed sandstones (Fig. 2: rock type RB) are quite uniform. The M_s(T)-curves of these rocks are predominated by a paramagnetic signal, but a weakly defined Curie point is detected at $680\,^{\circ}C$. The small increase in induced magnetization around $500\,^{\circ}C$ upon heating is ascribed to the destruction of clay minerals or micas, and to the formation of thermally unstable ferromagnetic phases (maghemite?). Saturation of IRM can not be achieved by fields up to 1 T. The coercivity spectrum is steadily distributed with high-coercivity parts (> 0.3 T) predominating. The remanence carrier of the red beds is attributed to haematite of variable grain size, but with a prevailing fine-grained fraction (pigmentary haematite). There is also evidence for coarse-grained specular haematite, because some samples have maximum unblocking temperatures of IRM between 600 and $675\,^{\circ}C$ when $> 30\%$ of magnetization is removed. The small low-coercivity component in the IRM acquisition curves (knee at 0.1 T) and the tiny inflexion near $600\,^{\circ}C$ during IRM thermal demagnetization point to magnetite as a mineral of minor significance in the unmetamorphosed red beds.

The metamorphism increases progressively, but not regularly towards the intrusion contact and leads to a drastic alteration of the ferromagnetic paragenesis over a short distance. Four magnetically different metasandstones (rock types A–D in Fig. 2) may be distinguished in the contact aureole. Because of the pronounced heterogeneity of the rock magnetic properties of the grey metasandstones, different magnetic characteristics are observed at the same locality.

Strongly metamorphosed rocks near the contact (rock type A at site 1) are characterized by low coercivity of IRM, being saturated at 0.2 T. The almost completely reversible M_s(T) curve shows a

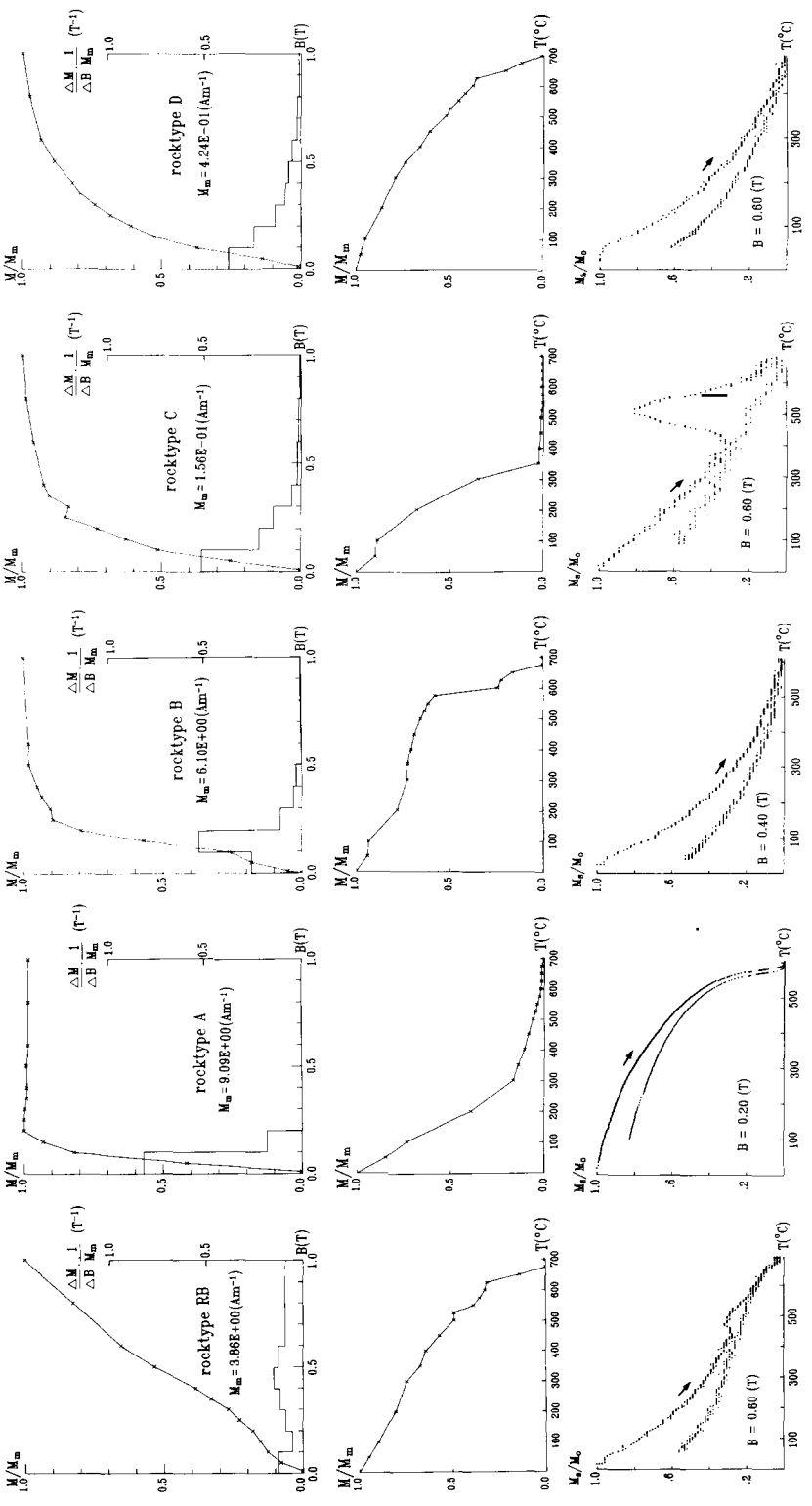


Fig. 2. Rock magnetic properties of Permian red beds (rock type RB) and metasandstones (rock types A–D). Upper row: IRM acquisition; middle row: IRM thermal demagnetization; lower row: strong field induced magnetization versus temperature, $M_s(T)$.

well-defined Curie point around 580 °C (Fig. 2). These properties are due to coarse-grained magnetite visible under the microscope. More than 75% of its saturation IRM is removed by demagnetization up to 300 °C. Instability of magnetite remanence is also indicated by a large NRM

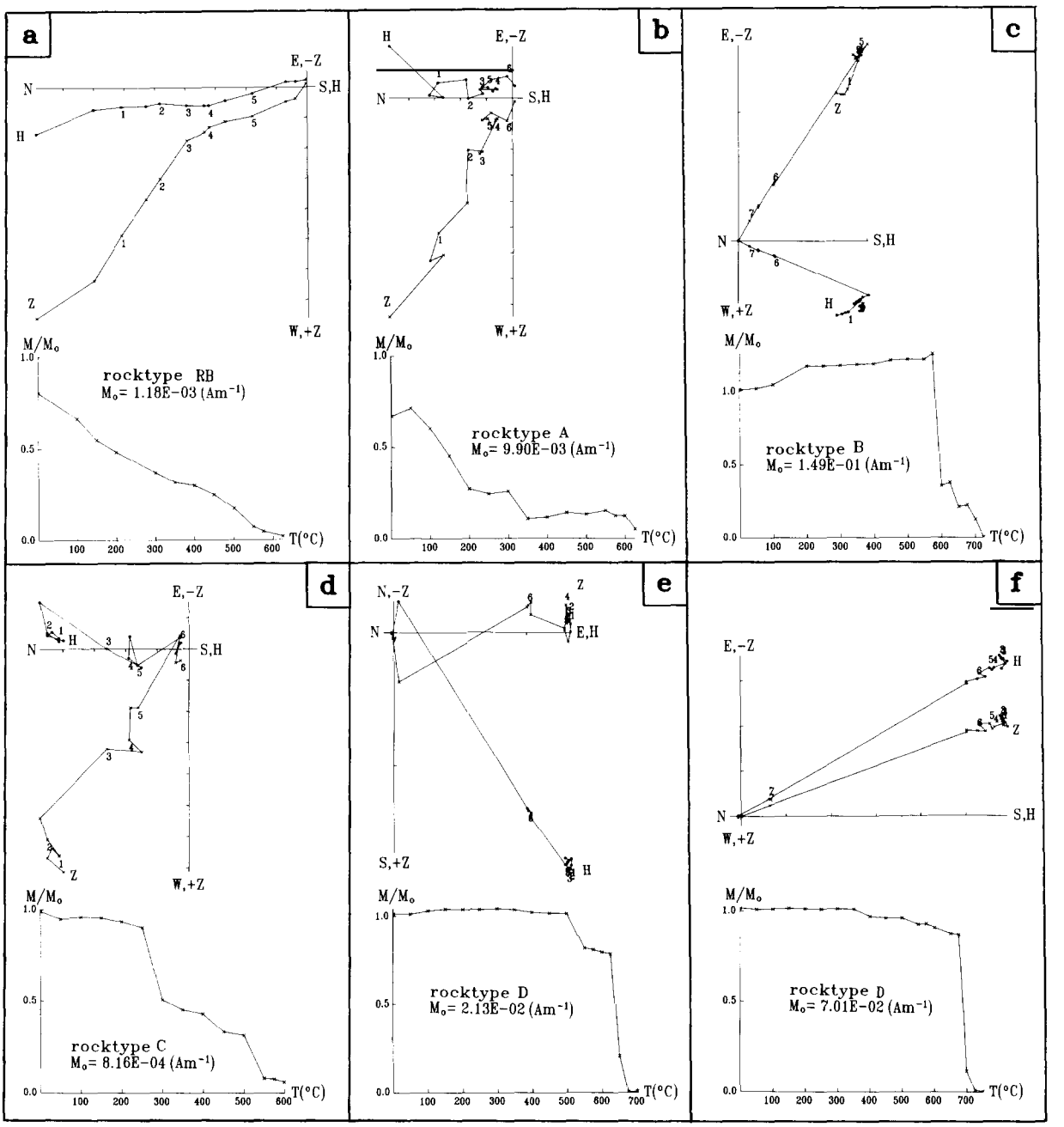


Fig. 3. Palaeomagnetic signature of (a) red beds and (b–f) metasandstones: NRM during thermal demagnetization. Horizontal (H, dots) and vertical (Z, crosses) NRM components for each sample have been plotted on orthogonal projection, where numbers denote temperatures divided by 100 °C. NRM intensity normalized versus room temperature NRM.

component acquired during a storage test over 3 months in the laboratory field. These short-lived viscous components amount to 30% of the initial NRM (Fig. 3b, rock type A). The magnetite phase, which does not contribute a stable natural remanence, has been formed during metamorphism. Magnetite formation in red sediments has been reported from laboratory experiments in air at temperatures $> 600\,^{\circ}C$ (Stephenson, 1967) and from baked sandstone inclusions in a basalt (Halvorsen, 1975). In the rock type A metasandstones, haematite contributes a very small IRM component which is demagnetized between 600 and 700 °C (Fig. 2), and carries the palaeomagnetically important, stable part of NRM (Fig. 3b).

The IRM coercivity spectrum of rock type B (Fig. 2) ranges upward to 0.5 T peaking between 0.1 and 0.2 T. The major proportions of IRM and NRM (Fig. 3c) are unblocked in two steps at discrete temperature intervals between 575 and 625 °C or between 650 and 700 °C. The NRM resides in high-coercivity material and is demagnetized only in alternating fields > 0.15 T. A weak paramagnetic signal characterizes the $M_s(T)$ behaviour. These properties are due to two types of haematite which have been identified optically. The lower unblocking phase is titanohaematite as a groundmass of exsolved haemoilmenites. The higher unblocking phase is either detrital specularite (martitized magnetite) or haematite formed during metamorphism. A few relics of magnetite sometimes contribute a small but unstable magnetization component.

The rock magnetic properties of rock type D (Fig. 2) are similar to those of rock type B. The paramagnetic $M_s(T)$ behaviour and the high coercivities and unblocking temperatures of IRM indicate again haematite as the major magnetic mineral. IRM (and NRM) intensities are generally lower by a factor of 10, and the shape of the IRM acquisition and thermal demagnetization curves is less square-shaped as that of rock type B. Remainders of pigmentary haematite, which give a faint reddish tint to some rock type D samples, are still contributing to IRM.

Iron sulphides control the magnetic properties of rock type C. The coercivity spectrum derived from IRM acquisition curves is basically concentrated in the field range below 0.5 T, but, always contains in addition, small high-field proportions (Fig. 2). The IRM is completely unblocked upon thermal demagnetization below 350 °C and so is much of the NRM (Fig. 3d). In the latter curve the remainder of NRM is removed below 575 °C, indicating the presence of a magnetite component. This component has probably been created during heating because often a strong increase of low-field susceptibility by several orders of magnitude is observed during heating above 400 °C, indicating phase changes such as conversion of pyrrhotite to magnetite (Schwarz, 1975). Phase changes are also recorded in $M_s(T)$ curves where irreversible changes of the mainly paramagnetic signal occur between 400 and 600 °C.

3. Palaeomagnetic results

The NRM intensities of the unmetamorphosed red beds group tightly around a mean of

TABLE I
NRM intensity, low field susceptibility and Koenigsberger factor Q (NRM versus induced magnetization)

Rock type	n	NRM (10^{-2} A m^{-1})	Susceptibility (10^{-4} SI units)	Q
RB	49	0.19 ± 0.19	1.20 ± 0.58	0.38 ± 0.15
A	16	5.4 ± 5.2	56.0 ± 46.0	0.22 ± 0.12
B	19	20.0 ± 9.17	1.92 ± 0.33	25.8 ± 10.6
C	26	0.29 ± 0.43	0.49 ± 0.39	1.23 ± 1.06
D	27	3.47 ± 5.65	1.63 ± 2.68	10.9 ± 17.6

1.9 mA m^{-1} and their Koenigsberger factors are low ($Q < 1$) (Table I, rock type RB). The corresponding parameters of the grey metasandstones depend strongly on the magnetic rock type as defined above. High low-field susceptibilities in rock type A are in accord with the presence of magnetite, whereas high Koenigsberger factors ($Q \gg 1$) support the evidence for haematite or titanohaematite in rock types B and D.

Directional analysis of the red bed NRM during thermal demagnetization shows that the vector usually contains three components. Viscous components which have been acquired in the laboratory field during a 3 month storage test, amount to ~ 25% of the initial NRM and indicate strong viscosity probably residing in the pigmentary haematite. Similarly, a noticeable soft component is removed during a 3 month storage in field-free space (first step in the demagnetization diagram of Fig. 3a: rock type RB). Up to 400 °C a direction with northerly declination and steep positive inclination is discerned. Several red bed samples (Table II; Sites 10, 13, 15) show only this direction throughout the whole range of demagnetization temperatures (up to 700 °C) and two samples carry this direction in the opposite sense (Fig. 4). The

TABLE II
Site mean directions of characteristic NRM

Site no. (rock type)	n	Temp. (°C)	Dec. (°)	Inc. (°)	Normal polarity (%)	α_{95} (°)	k
Metasandstone							
1 (A,B)	3	400–700	4.6	57.9	100	13.5	84.92
	2	550–650	322.3	14.0	100	–	–
2 (B)	9	500–700	211.1	−59.2	33	9.6	29.99
3 (C)	2	300–500	5.3	46.5	100		
4 (D)	7	500–650	149.8	−20.7	29	11.1	30.69
	3	650–700	6.9	43.4	100	18.8	44.16
5 (D)	6	500–700	150.9	−25.5	0	8.3	66.63
	1	650–675	299.0	68.8	100	–	–
Red beds							
6 (RB)	7	0–500	3.4	61.7	100	7.7	62.73
	6	500–700	313.7	21.8	100	20.2	11.95
7 (RB)	6	100–400	18.9	50.7	100	7.9	72.75
	6	400–650	332.2	25.8	100	14.2	23.11
8 (RB)	5	0–350	31.1	65.9	100	11.3	46.81
9 (RB)	5	0–550	25.7	57.5	100	6.1	158.64
	4	400–650	327.5	22.0	100	13.0	50.83
10 (RB)	6	0–700	10.0	54.2	83	8.7	60.26
	6	400–700	339.5	24.8	100	11.4	35.59
11 (RB)	1	50–250	336.1	68.6	100	–	–
	3	500–700	298.6	17.9	100	30.5	17.45
12 (RB)	3	50–300	338.6	65.2	100	9.6	166.61
	6	400–600	326.5	27.4	100	9.0	55.86
13 (RB)	5	0–650	16.8	54.1	100	29.3	7.77
	7	400–600	349.0	15.8	100	5.9	105.41
14 (RB)	3	50–350	348.8	67.5	100	7.4	275.40
	6	400–600	332.4	34.1	100	13.1	27.27
15 (RB)	7	50–650	6.6	68.1	86	11.7	26.54
	4	500–650	318.4	20.5	100	18.9	24.57

n: Number of samples; Temp: range of demagnetization temperatures of the respective component; α_{95}, k: Fisher (1953) statistical parameters

characteristic remanent magnetization (ChRM), which is carried very near the origin of the orthogonal vector diagram, is usually observed above 400 °C (Fig. 3a) and completely unblocked at temperatures well above 600 °C. This haematite component is characterized by shallow positive inclinations with declinations of 320–330 °.

Three NRM components can also be distinguished in the metasandstones. A fairly strong viscous magnetization which is removed during a

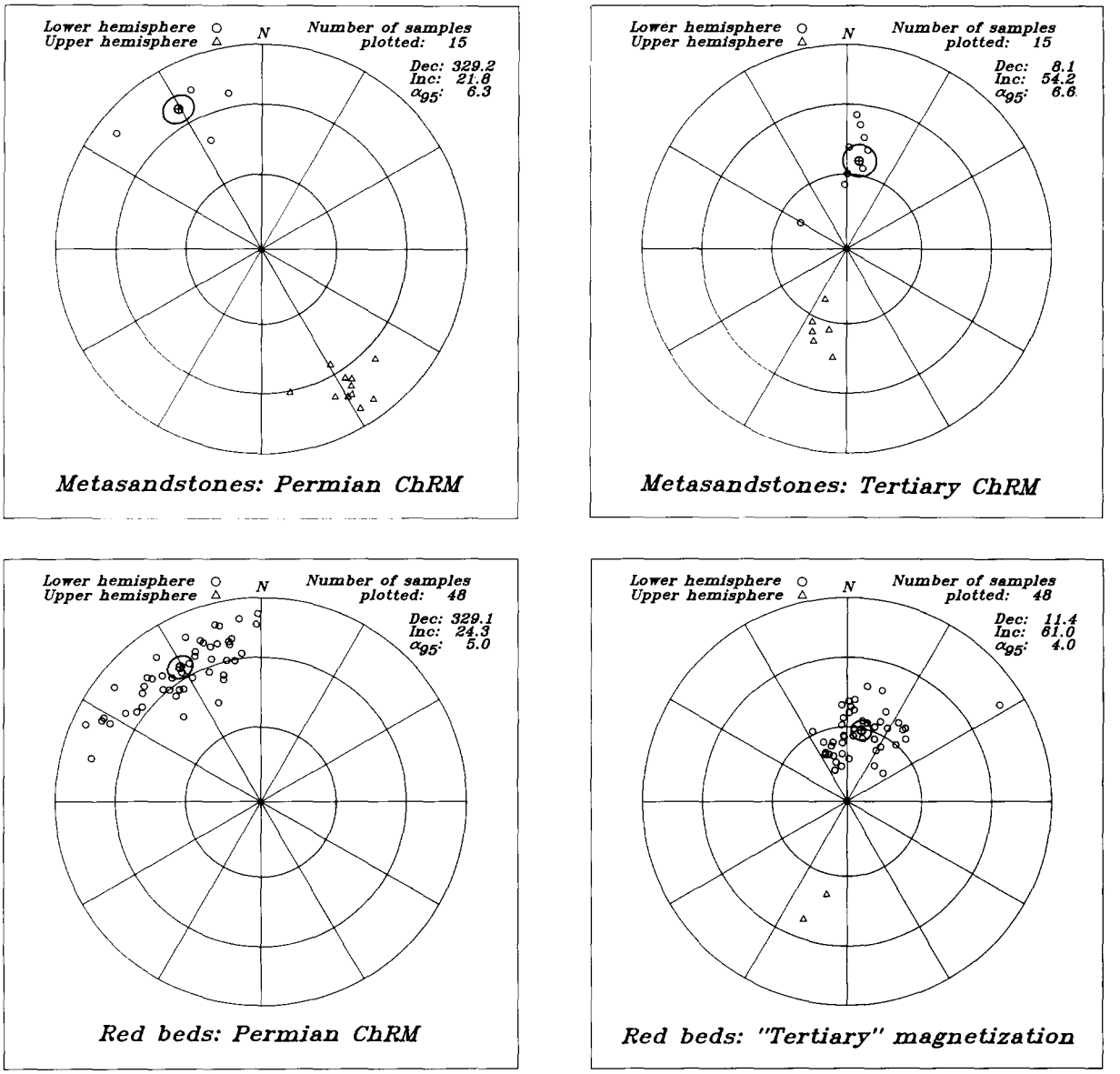


Fig. 4. Equal area stereographic projection of the Permian and Tertiary characteristic directions (ChRM) of red beds and metasandstones in the Val Daone (Adamello massif).

3 months storage test in field-free space, is always observed in rock type A (plotted again as first demagnetization step in Fig. 3b). Rock types B, C and D are not affected by VRM. The characteristic NRM directions observed during thermal demagnetization are similar to those measured in the unmetamorphosed red beds. Because of the different magnetic mineralogy, however, the intensity decay varies. Rock type A shows a relatively smooth decay with distributed blocking temperatures up to 400 °C where the magnetite remanence is removed and a low intensity plateau due to the presence of small amounts of haematite is reached. This high-temperature component has either a steeply inclined northerly or a shallow northwesterly direction (Table II). The ChRM components of rock types B and D residing in haematite are always unblocked in a very narrow discrete temperature interval, usually > 600 °C. The steep direction is preserved in rock type B, which is situated very near to the intrusion contact whereas shallow directions prevail in rock type D farther away from the contact (Table II; Fig. 1). The few samples of rock type C which carry a stable magnetization, contain the steeply inclined direction.

The resulting two characteristic mean directions (Fig. 4) can be compared with standard palaeomagnetic polar wander curves (Irving, 1977) and other palaeomagnetic data from the Southern Alps (Vandenberg and Zijderveld, 1982). The directions with shallow inclinations can be assigned a Permian age. The predominantly negative polarity in the metasandstones which contrasts the always normal polarity of the unmetamorphosed red beds, is explained best by the different stratigraphic position in the sedimentary sequence. The steeply inclined directions can be associated with the mid-Tertiary event of contact metamorphism, although viscous remagnetization in the present field cannot be entirely excluded.

The mainly inversely magnetized rock type B, however, points to a Tertiary origin of this magnetization component as a TRM or pTRM. The variable NRM polarity of rock type B (Table II: Site 2) may be interpreted in analogy to the titanohaematite-bearing granitic rocks of the Bergell intrusion as evidence for NRM self-reversal (Heller and Egloff, 1974). The steeply inclined magnetization isolated in some samples of rock type D at very high temperatures after the removal of a Permian direction at lower temperatures (Fig. 3e; Table II: Site 4) is understood only, if the formation of a new haematite phase and associated magnetization during Tertiary metamorphism is postulated.

The profile sampled across the contact zone can be subdivided in three different parts. The rear segment farthest away from the contact is represented by the unmetamorphosed to anchimetamorphic red beds. They contain a characteristic, but secondary, NRM component with moderately high unblocking temperatures related to the Tertiary metamorphism. The grain size of haematite carrying this component is probably small (pigment), whereas the ChRM fixed in larger grains with very high unblocking temperatures up to 650 °C manifests the Permian history of these rocks.

Reversed Permian ChRM directions are found in the grey metasandstones of rock type D in the middle part of the profile. In contrast to the red beds, the magnetization of these metamorphosed grey sandstones shows hardly any sign of Tertiary metamorphism, especially at lower demagnetization temperatures. Only Permian directions have been measured which are carried by specularite grains deposited during the original sedimentation in the Permian. Their blocking temperature spectra are restricted to values > 600 °C. The metamorphic event has leached the pigmentary haematite completely or the negligible pigment remainder has been reduced to superparamagnetic grainsizes and does not contribute to the NRM. In a few exceptional samples, an additional Tertiary component, which resides in haematite formed during the metamorphism, is recognized at even higher blocking temperatures than the Permian direction.

Most of the samples in the front of the profile up to 1.5 km from the intrusion contact (rock type A–C) carry Tertiary ChRM directions. Nevertheless, some Permian ChRM has been found even in the highest metamorphic rocks close to the intrusion boundary (Table II: Site 1). Two remanence carriers have to be distinguished: on the one hand,

a small stock of unremagnetized surviving Permian haematite, on the other hand a majority of new Tertiary magnetic minerals (magnetite, haematite, pyrrhotite) which probably preserve a partial or full thermoremanent magnetization due to the thermal event of the intrusion.

4. Discussion

An apparently paradoxical palaeomagnetic situation is encountered along the profile sampled towards the intrusion contact. The Permian magnetization of the red beds in the unmetamorphosed rear part of the section (Fig. 1: Sites 6–15) is overprinted to a variable degree by a secondary magnetization. Unblocking temperatures, usually up to 500 °C and sometimes even to 675 °C and occasionally reversed polarity of the NRM component, can be explained by Tertiary remagnetization causing a partial thermoremanent magnetization (pTRM) at elevated temperature during emplacement of the Adamello pluton and/or may result from burial beneath younger stratigraphic units (Kent, 1985). According to the model calculations of Pullaiah et al. (1975), temperatures of ~ 250 °C must have been reached. Similar temperature estimates have been obtained taking a 6 km Mesozoic sediment cover into account (Riklin, 1983). The completely metamorphic grey sandstones of rock type D in the middle part of the contact zone (Fig. 1: Sites 4,5), however, contain Permian directions only. Remagnetization due to Tertiary metamorphism has not been recorded. Tertiary directions again predominate in rock types A–C in the close proximity of the contact zone (Fig. 1: Sites 1–3), but part of the Permian magnetization has survived the intrusive heating even in the highest metamorphic sandstones (rock type A).

Red beds normally contain two types of haematite: coarse-grained specularite and fine-grained pigmentary haematite (Turner, 1980). The grain-size spectra may be continuous. Chemical demagnetization with hydrochloric acid has been used successfully in order to determine the haematite type which carries the characteristic NRM (Collinson, 1965; Park, 1970; Henry, 1979). The pigmentary haematite is dissolved first during leaching, while coarse-grained specularite survives the treatment. Collinson (1965) showed that most of the characteristic stable NRM of certain red beds is carried by specularite. In contrast, other studies point to pigment haematite as the main NRM carrier (Collinson, 1974; Turner, 1980).

The Tertiary metamorphism of the Verrucano Lombardo red beds due to the Adamello intrusion is a unique example of natural chemical demagnetization. The progressive influence of the contact metamorphism changes the rock colour from red to grey. The discoloration indicates complete destruction of the pigmentary haematite at distances < 2 km from the contact (i.e., in the middle part of the palaeomagnetic profile). It is evident from simple thermodynamical considerations that mineral phases having a large surface and a small volume are more reactive than phases characterized by a large volume and a relatively small surface area. Thus the fine-grained haematite has preferentially been the source material for new micas to form in the biotite metamorphic zone while coarse-grained haematite remains in the rock. The grain-size spectrum shifts to preference of larger volumes which have lower coercive forces and discrete high blocking temperatures (Fig. 2).

Consequently, secondary NRM components with low to moderate blocking temperatures can not be imprinted during metamorphism. The Tertiary component is missing in rock type D and the original Permian specularite stock controls the NRM of these rocks. In the unmetamorphosed red beds, however, the overprint component, if VRM or pTRM, resides in the haematite pigment. Since the blocking temperature spectra are changed completely and restricted to high temperatures in this way, it is not possible to estimate the temperature distribution in the zone of contact metamorphism from unblocking of the NRM (Dunlop and Buchan, 1977).

The high-grade metamorphism causes minerals to be depleted in ions which do not fit well in their crystal structure. The available Fe may form new Fe minerals, such as Fe sulphides, magnetite or even haematite. The new magnetic phases are expected to be magnetized along the direction of the ambient field during their formation, i.e., the

direction of the Tertiary geomagnetic field. In fact, Tertiary magnetizations are by far the most prominent magnetization components observed in rock types A–C (Table II). But a few examples of high-grade metamorphic rocks (such as rock type A) still contain original coarse-grained haematite with a Permian remanence. Survival of such old magnetization seems to be possible only in a metamorphic temperature regime which in accordance with petrological temperature estimates (Riklin, 1983) did not appreciably exceed temperatures of ~ 600 °C near the contact.

A northwestern declination (320–330 °) and a shallow inclination (< 30 °) characterize the old ChRM in the red beds of the rear end of the profile (rock type RB) and in the grey sandstones in the middle part of the contact zone (rock type D; reversed polarity). The same direction has been found in a few samples of the high-grade metamorphic rocks of rock type A. The mean direction (declination $D = 328.1$ °, inclination $I = 24.8$ °, $\alpha_{95} = 5.2$ °) agrees well with already published data for the Southern Alps (see summary in Haag, 1985). The result confirms the anticlockwise rotation of the Southern Alps since the Permian based on the apparent polar wander curve of the Southern Alps by Vandenberg and Zijderveld (1982). In contrast to observations from the Central Alps (Heller, 1980) the Tertiary directions reported here do not indicate rotational movements of the Southern Alps relative to Europe since the Oligocene.

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